

Homework 5: Theory

Due: July 19, 2026

Modern Computer Vision, Technion, Spring 2026

Did you ever think you'd need to solve a physics question in an AI course?

A particle in 1D takes i.i.d. steps $\pm a$ by a fair coin flip.

- (a) Let $\rho(x, t)$ be the PDF of the particle's location. Show that X_N , the location after N steps, satisfies $X_N \approx \mathcal{N}(0, Na^2)$ for large N .
- (b) Deduce $\mathbb{E}[X_N^2] = 2Dt$ at time $t = N\tau$ (with τ the time per step), and identify D in terms of a and τ . Why is holding a^2/τ fixed the *only* scaling that survives the limit? (*What happens to the spread otherwise? Check both directions.*) This $\sqrt{t}$ spread is the Einstein–Smoluchowski relation.
- (c) Show a step convolves the density with the step law k :

$$\rho(x, t+1) = (\rho(\cdot, t) * k)(x), \quad k(u) = \frac{1}{2}\delta(u-a) + \frac{1}{2}\delta(u+a).$$

- (d) Take the continuum limit of (c) to show

$$\frac{\partial \rho}{\partial t} = D \frac{\partial^2 \rho}{\partial x^2}, \quad D = \frac{a^2}{2\tau}.$$

(*Taylor-expand $\rho(x \pm a)$ to order a^2 and average with weights $\frac{1}{2}, \frac{1}{2}$; the odd term cancels. Subtract ρ , divide by τ , and send $a, \tau \rightarrow 0$ with a^2/τ fixed.*) This is the heat equation.

- (e) Find the velocity field $v(x, t)$ for which the heat equation (d) can be written as a transport (continuity) equation,

$$\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial x}(\rho v).$$

(*Rewrite the right-hand side of (d) as $\partial_x(D \partial_x \rho)$ and match it to the form above; the identity $\partial_x \rho / \rho = \partial_x \log \rho$ will help simplify.*) The v you find is the velocity that transports the density — it is the *score* that every diffusion model learns.

- (f) (*Bonus.*) Repeat (c)–(d) with an unfair coin: $+a$ with probability p , $-a$ with $1-p$ (a particle on a uniform slope). (*The odd term that cancelled in (d) no longer does — it survives as a new first-order term.*) What equation does the density obey now? The result is the Fokker–Planck equation (the fair coin is its driftless case).

Optional. Check by substitution that the drifting Gaussian

$$\rho(x, t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{(x-vt)^2}{4Dt}\right)$$

solves it. What does v do to the bell curve, and what does D do?